NutriPantry: Database Design Document

DBMS Experiential Learning Project

ER Diagram and Relational Schema

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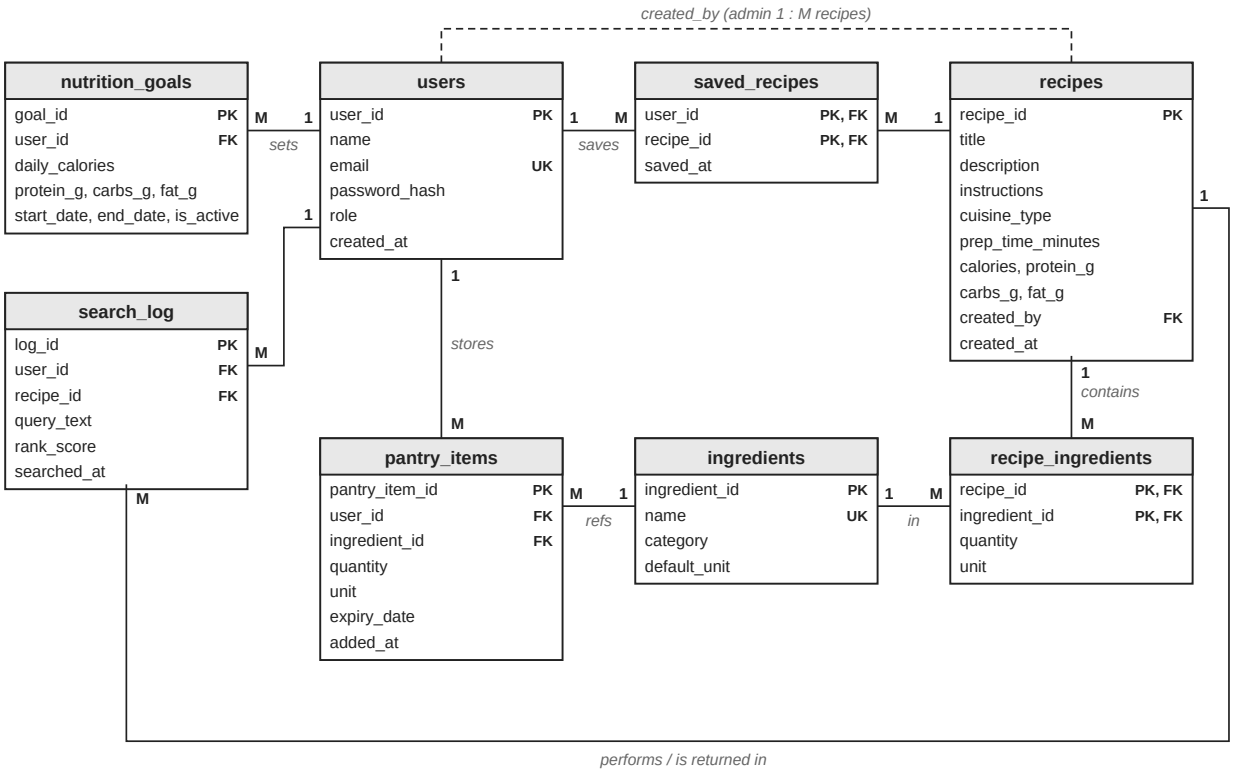
The structured data is stored in PostgreSQL across eight tables, two of which are junction tables (`recipe_ingredients` and `saved_recipes`). Recipe text is embedded separately in a vector store for semantic search.

Entity-Relationship Diagram



PK: primary key. FK: foreign key. UK: unique constraint. Dashed line: admin-to-recipe ownership through `created_by`.

Relationships

A user owns many pantry items, nutrition goals and search log entries. Pantry items reference the master ingredient list. Users and recipes are linked many-to-many through `saved_recipes`, and recipes and ingredients through `recipe_ingredients`. Recipes are created by admin users through `created_by`.

Relational Schema (DDL)

`users`

Account and role.

```
CREATE TABLE users (
    user_id       SERIAL PRIMARY KEY,
    name          VARCHAR(100) NOT NULL,
    email         VARCHAR(150) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    role          VARCHAR(20) NOT NULL DEFAULT 'user'
                  CHECK (role IN ('user', 'admin')),
    created_at    TIMESTAMP NOT NULL DEFAULT NOW()
);
```

`ingredients`

Master ingredient list.

```
CREATE TABLE ingredients (
    ingredient_id SERIAL PRIMARY KEY,
    name          VARCHAR(100) UNIQUE NOT NULL,
    category      VARCHAR(50),
    default_unit  VARCHAR(20) NOT NULL
);
```

`pantry_items`

What each user has on hand.

```
CREATE TABLE pantry_items (
    pantry_item_id SERIAL PRIMARY KEY,
    user_id        INT NOT NULL REFERENCES users(user_id)
                   ON DELETE CASCADE,
    ingredient_id  INT NOT NULL REFERENCES ingredients(ingredient_id),
    quantity       DECIMAL(8,2) NOT NULL CHECK (quantity >= 0),
    unit           VARCHAR(20) NOT NULL,
    expiry_date    DATE,
    added_at       TIMESTAMP NOT NULL DEFAULT NOW(),
    UNIQUE (user_id, ingredient_id)
);
```

`nutrition_goals`

Per-user daily targets.

```
CREATE TABLE nutrition_goals (
    goal_id        SERIAL PRIMARY KEY,
    user_id        INT NOT NULL REFERENCES users(user_id)
                   ON DELETE CASCADE,
    daily_calories INT NOT NULL CHECK (daily_calories > 0),
    protein_g      DECIMAL(6,2),
    carbs_g        DECIMAL(6,2),
    fat_g          DECIMAL(6,2),
    start_date     DATE NOT NULL,
    end_date       DATE,
    is_active      BOOLEAN NOT NULL DEFAULT TRUE
);
```

recipes

Recipe catalog with nutrition facts.

```
CREATE TABLE recipes (
    recipe_id         SERIAL PRIMARY KEY,
    title             VARCHAR(150) NOT NULL,
    description       TEXT,
    instructions      TEXT NOT NULL,
    cuisine_type      VARCHAR(50),
    prep_time_minutes INT CHECK (prep_time_minutes > 0),
    calories          INT,
    protein_g         DECIMAL(6,2),
    carbs_g           DECIMAL(6,2),
    fat_g             DECIMAL(6,2),
    created_by        INT REFERENCES users(user_id),
    created_at        TIMESTAMP NOT NULL DEFAULT NOW()
);
```

recipe_ingredients

Junction table: recipe to ingredient (many-to-many).

```
CREATE TABLE recipe_ingredients (
    recipe_id     INT NOT NULL REFERENCES recipes(recipe_id)
                  ON DELETE CASCADE,
    ingredient_id INT NOT NULL REFERENCES ingredients(ingredient_id),
    quantity      DECIMAL(8,2) NOT NULL,
    unit          VARCHAR(20) NOT NULL,
    PRIMARY KEY (recipe_id, ingredient_id)
);
```

saved_recipes

Junction table: user to favourite recipe (many-to-many).

```
CREATE TABLE saved_recipes (
    user_id   INT NOT NULL REFERENCES users(user_id)
              ON DELETE CASCADE,
    recipe_id INT NOT NULL REFERENCES recipes(recipe_id)
              ON DELETE CASCADE,
    saved_at  TIMESTAMP NOT NULL DEFAULT NOW(),
    PRIMARY KEY (user_id, recipe_id)
);
```

search_log

Recommendation query history.

```
CREATE TABLE search_log (
    log_id      SERIAL PRIMARY KEY,
    user_id     INT NOT NULL REFERENCES users(user_id)
                ON DELETE CASCADE,
    recipe_id   INT REFERENCES recipes(recipe_id)
                ON DELETE SET NULL,
    query_text  TEXT NOT NULL,
    rank_score  DECIMAL(6,4),
    searched_at TIMESTAMP NOT NULL DEFAULT NOW()
);
```